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An Independent Problem

Is it possible to be certain that I'm not dreaming now? This question is unsolvable because this type of problem is an Independent Problem. An independent problem is a problem that is independent of the system of logic used to solve the problem. Descartes illustrates this with his received view of dreaming argument. Therefore, you cannot know if you are not dreaming now.

Descartes introduces this question with his dream argument. First, he begins by asking himself what he can know for certain, and that is that he is currently sitting in front of a fire. However, he realizes he actually can't know that because he has performed this same action before thinking it was reality only to wake up from a dream. He extends this concept to be universally true for all dream and non-dream state conditions.

Moreover, Descartes uses what is now called the "received view of dreaming" argument to affirm his reality. Descartes argues that reality and dreams occur in a sequential order, not in a flash. He goes on to say that events occurred in a dream are the only events remembered and events that didn't occur are not remembered. For instance, if a person has a dream containing events A through G, then those are the only events that are remembered. Events Y through Z are not remembered because they didn't occur in the dream. Therefore, it's impossible to remember events that didn't take place when dreaming.

I posit the following statement that all elements in reality must be present in a dream, at the bare minimum, to determine the experience is in fact reality. This is supported by Descartes' received view of dreaming argument because all elements in a dream come from reality thereby making the elements in a dream a subset of the elements in reality.

Determining this in and of itself may seem like an impossible task. Fortunately, mathematicians have come up with a way of classifying such tasks. First, we must follow the Tarski-Kuratowski algorithm and state the problem in prenex normal form. This would result in the following: for all elements in reality, there exists an element in this dream such that the element in this dream is equal to the element in reality. Or more formally:

Let r be an element of true reality, d be an element of the currently experienced reality, and $\psi(r,\ d)$ be a comparison between r and d that returns true if and only if r and d are equivalent by every known means:

$$\forall r \exists d(\psi\left(r, d\right))$$

Given this as the result we begin with a universal quantifier followed by an existential quantifier and a quantifier free matrix. This places the complexity class into

Π_2 in the arithmetical hierarchy. Other problems in this classification include the famous twin primes conjecture, which was proposed in 1846 and has remained unsolved. Perhaps the most damning problem is the complement to the halting problem which is known to be undecidable due to its independence. It looks like then at best the problem of determining that we are not in a dream is undecidable.

Solving the inverse for the prompt is not nearly as difficult as solving for the prompt itself. This means that it is easy to know that you are in a dream compared to knowing that you are not in a dream. A perfect example is lucid dreamers who practice this on a regular basis. Other problems in Π_2 share similar characteristics. The complement to the halting problem is a great example to compare this problem to.

In theoretical computer science, the halting problem asks if we can determine whether or not an arbitrary program will eventually finish running or halt. For programs that do halt., the answer is yes, we can know that they halt. This is quite similar to the fact that we can know that we are dreaming.

The "complement to the halting problem" is simply the inverse of this – given an arbitrary program, will it run forever? We can know for sure when this is false, i.e. when the program does halt. However, for programs that do not halt, we cannot develop any general method to know that they do not halt. The problem is *undecidable*. The problem of determining that we are not dreaming is also a decision problem – it is a true or false and is likewise undecidable.

This is why the classification of Π_2 in the arithmetical hierarchy is useful. Problems in Π_2 can be shown to be true, often trivially, but may be undecidable if they are false. As we have reformulated the question as "does this reality have all elements of the 'true' reality", we end up with a useful result showing the semi-decidability of the problem.

Now that we have shown that our "subproblem" is undecidable, can we take this further and show that the problem of determining that we are not dreaming is independent? We already know that we cannot know that we are not dreaming, as, at a bare minimum, we would have to decide an undecidable problem. By revisiting the received view of dreaming, we can determine that the problem is even more difficult – it is not just undecidable but independent.

According to the received view of dreaming, every element in a dream is "received" from a "higher" reality. If this is the case, then every dream, at best, can contain every element of the higher reality, and is also capable of only containing a subset of the elements of the higher reality. We already know that only dreams containing all elements of reality are candidates for being reality, and that determining whether or not this is the case is undecidable.

What if we are missing elements from a higher reality? What would those elements look like? These elements, by definition, would be unknowable in the dream. Even the capability of thinking about these elements would prove their existence within the dream. Therefore, if there are elements of reality that are missing from our current dream, by definition, we cannot even conceive of or think about these missing elements!

Let's imagine for a moment that we are dreaming, and not in "true" reality. How would we know if elements of reality are missing? Even if we had a guide that could perfectly describe every missing element, we should not be able to understand such a guide, as its own existence would mean that the elements of reality would be brought into our dream. Therefore, when in a dream, some elements of reality are logically independent from the dream.

Now let's imagine we are in reality. How would we know? Even if every element of reality is present, we cannot perceive that to be the case. Our own human interpretation of the world contains only a subset of the true reality and is therefore akin to a dream. We already showed that we cannot bring all elements of reality into a dream. Therefore, even if we are living in "reality", the problem of determining that we are not dreaming is still not only undecidable, but independent.